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6-DOF object tracking method based on lensless event sensor

Yue You, Mingzhu Zhu *, Hong Zhang, Bingwei He

School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou 350108, China

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ABSTRACT

Event sensors can capture fast-moving objects without motion blur, but their limited spatial resolution constrains tracking accuracy. To address this issue, we propose a method for achieving precise 6-DOF object tracking using lensless event sensors. A mask printed with marker fields replaces traditional lenses, and a component composed of multiple rigid beacons is used to mark the object. Data deconvolution and matching optimization algorithms are developed to calculate the positions of multiple beacons and estimate the object's pose, employing a dual-filter to smooth the tracking trajectory. Experimental results demonstrate that, compared with state-of-the-art methods, our method reduces positional error by 19.6%, from 1.68 cm to 1.35 cm, while achieving rotation accuracy, 2.94° versus 3.08°, at a tracking frequency of 500 Hz. Additionally, it maintains low computational cost with an average processing time of 1.71 ms per frame, thereby satisfying the requirements of real-time applications.

1. Introduction

Visual-based object tracking method has been widely applied in the fields of virtual reality [1], intelligent monitoring systems [2], vibration measurement [3], and medical imaging [4]. They mostly use Charge Coupled Device or Complementary Metal Oxide Semiconductor (CCD/CMOS) sensors to obtain object information. However, these sensors capture the light intensity of all pixels simultaneously at a fixed sampling rate. This working principle leads to motion blur when tracking fast-moving objects, leading to reduced accuracy. Even if high-speed cameras can be used to increase the sampling rate, the corresponding large amount of data imposes a burden on subsequent computations.

As another type of visual sensor, event sensors [5] compensates for the shortcomings of CCD/CMOS sensors, as they only capture light intensity changes as events consisting of pixel coordinates, timestamps, and polarities. These events are asynchronously recorded and transmitted, without waiting for the completion of light intensity collection across all pixels. This operational principle enables them to achieve a microsecond-level time resolution, effectively eliminating motion blur. Additionally, the output data is presented as quadruples, reducing data-throughput and computational-complexity. However, event sensors have their own drawbacks. The captured data does not directly represent grayscale values, thus failing to achieve sub-pixel tracking accuracy. Moreover, event sensors face limitations in rapidly processing a large number of events, thus their spatial resolution is manufactured to be low. This results in considerable errors in 6-DOF pose estimation based on event sensors, with position errors exceeding 3 cm and rotation errors exceeding 6 degrees [6]. Even the best-performing method reports position errors greater than 1.6 cm and rotation errors above 2.9 degrees [7]. Therefore, it is necessary to develop an event sensor based tracking method aiming to maintain high temporal resolution and improve accuracy. Moreover, lens-based event sensing systems are inherently bulky due to their optical components, which limits their applicability in size-constrained or weight-sensitive applications. Recent research has found that lensless systems can enhance tracking accuracy and overcome the aforementioned limitations. These systems use beacons to control the imaging process, generating chaotic images through specific media and linking the beacons' positions to the entire image. The slight motion of the beacons causes significant changes in the image, amplifying their motion and thereby improving tracking accuracy. However, the reliance on beacons may introduce certain limitations, which can be categorized into three main aspects: 1. Scalability. Beacons are unsuitable for open or large-scale environments, thereby limiting the scalability of this approach. 2. Deployment complexity. The physical installation and calibration of beacons often require manual intervention, increasing setup complexity. 3. Signal interference. Environmental factors such as occlusions and obstacles may degrade signal quality and localization accuracy. Despite these challenges, the proposed approach remains highly versatile and applicable across a wide range of scenarios [8–10].

Inspired by this, we propose an object tracking method based on lensless event sensor, which is capable of achieving high-precision

* Correspondence to: No. 2 Wulongjiang North Avenue, Fuzhou University City, Fuzhou City, Fujian Province, China.
E-mail addresses: yueyoufuz@163.com (Y. You), mzz@fzu.edu.cn (M. Zhu), zhngfzu@163.com (H. Zhang), mebwhe@fzu.edu.cn (B. He).

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tracking with minimal computational power. A mask printed with marker field [11] is placed in front of the event sensor, and the lights from the beacons are screened by the mask to form position-sensing information (see Fig. 1). Given that the event sensor only responds to changes in light intensity, beacons flicker at a fixed frequency. The proposed method utilizes data deconvolution and matching optimization to simultaneously localize these beacons. Regarding the selection of the number of beacons, fewer than three beacons result in the failure of attitude estimation, while more than three beacons increase the computational burden. Therefore, this paper demonstrates the feasibility of simultaneous positioning with three beacons, achieving a balance between computational efficiency and the threshold required for attitude estimation. In addition, to mitigate the singularities in tracking results caused by overlapping projections of multiple beacons, we design a dual-filtering approach, which integrates the Particle Filter (PF) [12] with the Unscented Kalman Filter (UKF) [13]. The contributions of this work are as follows: **1. System architecture design.** The proposed design, which replaces traditional lenses with mask plates, overcomes the limitations of conventional optical systems. In comparison to traditional methods, our approach significantly reduces hardware complexity while enhancing positioning accuracy. **2. Algorithm optimization.** The introduced data deconvolution, matching optimization, and dual-filtering algorithms not only enhance tracking accuracy but also improve the real-time performance of the system, making it more effective in high-dynamic environments. **3. Practicality and application prospects.** This method can effectively cope with scenarios with limited computing power and storage, especially in fields such as industrial automation, surgical navigation and augmented reality. With the improvement of spatial resolution of event sensors, we expect to further improve positioning accuracy. This work has the following advantages:

1. **High accuracy.** This method can achieve high-accuracy 6-DOF tracking, demonstrating significant improvements over traditional event-based systems, reducing positional errors by 19.6% and achieving rotation accuracy comparable to existing techniques.
2. **High frequency.** This method has low computational power requirements, and the frequency of pose estimation can reach 500 Hz, making it suitable for real-time tracking of fast-moving objects.
3. **Hardware simplified.** This method has the advantages of low cost, lightweight, small size, and manufacturing effective, making it more convenient to use in systems which are compact.

2. Related work

Event sensors [5] are introduced relatively late compared to more mature CCD/CMOS sensors, and their application in object tracking has only emerged recently. Nevertheless, due to their numerous advantages, many methods have been proposed.

In order to utilize the high temporal resolution of event sensors, tracking methods need to be fast and reliable. Earlier methods treat objects as clusters and update the cluster center by adding new events. Owing to their high computational efficiency and event-driven nature, these methods can accommodate high temporal resolution, making them well suited for tracking high-speed objects. For example, Aladem and Rawashdeh [14] propose a multi-cluster tracking method that fits the binary Gaussian distribution to each event cluster. The cluster center is determined by the Gaussian center, while the clustering direction is derived from the Gaussian covariance. Using a simple weighting strategy, the Gaussian distribution is updated based on nearby new events. Similarly, Dietsche et al. [15] propose a power line tracking method employing drones. This method generates event clusters with linear shapes to match the appearance of power lines. The decision

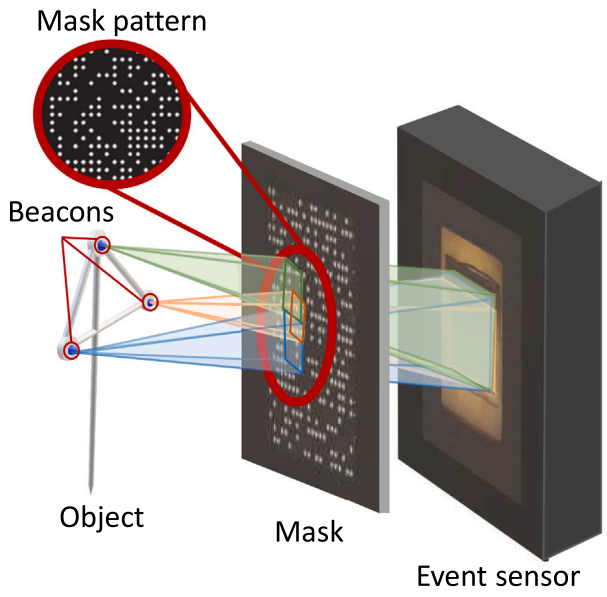


Fig. 1. System setup: Lensless system that consists of an event sensor and a mask painted with mark field.

to add an event to the cluster is made by assessing the distance from the power line. However, clusters are updated using simplistic spatial criteria. Any event close enough can affect the updating process, thus the noisy events produced by a moving background can smear and even overwrite the clusters.

Subsequent studies propose converting events within a given time window into feature representations and optimizing feature matching between consecutive time intervals to achieve object tracking. While these approaches increase computational complexity, they enhance the accuracy and robustness by suppressing background noise. As a result, they are suited for demanding scenarios, such as high-precision tracking and environments with drastic illumination changes. For instance, Liu et al. [16] propose a 6-DOF tracking method based on line features. Their method extracts object contours from events and refines the pose through event-line matching optimization by minimizing the distance between event points and lines. Subsequent improvements [17] involve detecting corners from the event-based time surface and computing their correspondences by maximizing the probability of temporal event correlation. The pose is further refined by minimizing the distance between matched corners and edges through optimization, enabling continuous tracking. Further developments [18] employ stereo event cameras to extract line features from the event stream, and reconstruct a wireframe model of the tracked object (e.g., a spacecraft). By formulating event-line matching as a continuous optimization problem, these methods achieve higher precision in pose tracking. However, all event-based sensors must process large volumes of events within short time intervals, which constrains their spatial resolution to tens of thousands of pixels. Although these methods offer improved accuracy compared to cluster-based tracking approaches, the absolute accuracy remains limited.

To enhance accuracy and robustness beyond basic object tracking performance, researchers incorporate motion filtering techniques based on various algorithms. Mitrokhin et al. [19] estimate motion parameters by minimizing the gradient difference between the event count image and the time surface image. Independent moving objects are detected based on inconsistencies between the temporal image and the global motion model, and a Kalman filter with a constant acceleration model is applied to generate tracking trajectories. Similarly, Li et al. [20] propose a filtering framework that leverages correlation

modeling to predict the motion offset of 2D features and the spatial position of 3D features. Motion filter is introduced to mitigate the effects of feature deformation, and a block matching module is integrated to estimate the object pose.

Recent research has found that lensless systems can enhance tracking accuracy in cases of limited sensor resolution. In such systems, the position information of a beacon is distributed across the entire sensor surface, enabling overdetermined localization and thereby enhancing tracking precision. A data deconvolution process is then applied to recover the beacon information. For example, Cai et al. [21] modulate light using a coded mask mounted on a lensless camera and employ spatially varying deconvolution to accurately reconstruct the range space. To further enhance resolution, a conditional diffusion model is introduced to supplement high-frequency details, achieving high-quality reconstruction. Similarly, Zhou et al. [22] utilize a point spread function in conjunction with a three-dimensional deconvolution algorithm to suppress imaging artifacts. A regularization strategy is also incorporated to balance artifact suppression and the preservation of weak signals, enabling adaptability to samples with varying densities. Inspired by this, we install a mask in front of the event sensor and propose the corresponding object tracking algorithm to simultaneously track three beacons, thereby achieving 6-DOF tracking (see Fig. 1).

3. Methods

3.1. System architecture

3.1.1. Mask pattern

The mask pattern is based on the De-Bruijn sequence [23], which is a sequence of k different symbols, and each subsequence of length n appears once in the entire sequence. For the proposed method, the binary De-Bruijn sequence ($k = 2$) is used, where the symbols are limited to 0 and 1. For instance, with $k = 2$ and $n = 2$, it is represented as [00 01 10 11]. Expanding the De-Bruijn sequence into two dimensions gives rise to the De-Bruijn Torus (DBT), denoted as $\{r, s, m, n\}$. Here, $r \times s$ represents the size of the DBT, and each $m \times n$ sized window appears exactly once in the entire DBT. This uniqueness is beneficial for determining the position of beacons.

For high tracking accuracy, the mask pattern with high dense information is required, but excessive density may introduce significant noise due to light diffraction. Hence, careful selection and processing of the DBT pattern are essential. A DBT of $\{256, 256, 4, 4\}$ is generated empirically with the algorithm of Schusselbauer et al. [24]. In order to enhance the clarity of the mask pattern, we convert the binary DBT 0/1 matrix into a dot matrix pattern. Here, we depict 1 as a small dot, while leaving the positions of 0 unprocessed, as illustrated in Fig. 1. To avoid insufficient dots in local patterns, we chose regions corresponding to the 128 × 128 DBT with a uniform dot distribution to create the dot matrix pattern.

3.1.2. Event sensor principle

Event sensors operate on a distinctive principle for data generation. As shown in Fig. 2(a), event sensors detect variations in light intensity at the pixel level. Once the change in its logarithmic surpasses a predefined threshold, an event is triggered, as described by:

$$|log[I(x, y, t)] - log[I(x, y, t - \Delta t)]| > \xi, \tag{1}$$

where $I(x, y, t)$ is the light intensity value in pixel (x, y) at time t, and Δt denotes a time interval unit measured in microseconds. The threshold is denoted as ξ. The resulting event $e = (x, y, t, p)$ is a quaternion. In this notation, (x, y, t) specifies the pixel's location and the timestamp at which the event occurs, while $p \in \{-1, 1\}$ represents the event's polarity, indicating a decrease or increase in light intensity. Arranging and transmitting events according to the timestamps is called an event stream.

3.2. Beacon projection positioning

Event sensors inherently introduce front-end noise due to thermal fluctuations in their circuitry, which manifests as irregular isolated activations in the event stream. To address this issue, spatial filters are employed to suppress such noise [25]. In addition, the flashing of beacons can produce a large number of spatially consistent events, which reduces algorithmic efficiency. Therefore, we sparse the event stream to reduce redundancy and improve processing efficiency. After these preprocessing steps, we can more effectively extract event information related to the object. It is needed to simultaneously determine the projection positions of three beacons in the mask to achieve object pose estimation. To address this, we extract information from the event stream by intercepting it using the time window and generating binary images. In these images, pixels containing events are represented as white (1), while other pixels are depicted as black (0). To enhance image clarity, we apply morphological opening and closing to eliminate small noise artifacts, as shown in Fig. 2(b). The morphological opening operation utilizing circular structuring elements effectively suppresses small-scale noise components within the image while preserving the essential features. Subsequently, a closing operation is applied to fill minor gaps and discontinuities, while maintaining the structural integrity of the original pattern. This combination method has low computational cost while making the beacon projection pattern clear [26,27]. We define these images as the simulated pattern I_s. The mask pattern, which is to be described in Section 3.1.1, is defined as I_m. However, the projections of the three beacons overlap, making it difficult to distinguish which signal corresponds to each beacon. To address this challenge, we propose a data deconvolution algorithm inspired by classical signal separation techniques in image processing [28,29], but specifically adapted to accommodate the unique characteristics of our beacon projection system. A cost function is introduced, as:

$$F_t[x, y] = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} I_s[i, j] \wedge I_m[i + x, j + y], \tag{2}$$

where $\wedge$ denotes element-wise logical AND. For each pixel, if both values equal 1, the result increments the total count by one. Thus, $F_t[x, y]$ serves as a metric to quantify the similarity between I_s and I_m. The dimensions of I_s are $m \times n$ pixels, and $M \times N$ pixels for I_m. Thus, $x = 0, 1, \ldots, M - m + 1$ and $y = 0, 1, \ldots, N - n + 1$ represent the valid ranges for computing the metric. A higher value of $F_t[x, y]$ indicates a greater resemblance. The heatmap for $F_t[x, y]$ is shown in Fig. 2(c).

The three local maximum regions within $F_t[x, y]$ are identified, introducing a constraint to ensure a local maximum:

$$\begin{aligned} &\{(x_1, y_1), (x_2, y_2), (x_3, y_3)\} = \arg\max_{x,y} \operatorname{top} 3 F_t[x, y] \\ &\text{s.t. } F_t[x_q, y_q] \geq F_t[x, y], \ \forall\ (x, y) \text{ for } \ q = 1, 2, 3 \\ &\text{in } \{(x, y) | \sqrt{(x - x_q)^2 + (y - y_q)^2} \leq \epsilon, \ (x, y) \neq (x_q, y_q)\}, \end{aligned} \tag{3}$$

where $\{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}$ represents the most probable 3 positions of the projection centers, as shown in Fig. 2(c), ϵ represents the radius size of the local maximum region.

However, $\{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}$ are just rough values, and a matching optimization algorithm is proposed to determine the precise values. The search window I_{wq} is built centering (x_q, y_q). The scaling ratio for the projection pattern to the mask pattern is denoted as s_q, and the corresponding search window I_{wq} can be expressed as:

$$\begin{aligned} &I_{wq}[i, j] = [\mathbf{T}_{\text{trans}}^{-1} \cdot \mathbf{T}_{\text{scale}} \cdot \mathbf{T}_{\text{trans}} \cdot I_m(i, j)]^{\top}, \\ &\text{where} \quad \mathbf{T}_{\text{trans}} = \begin{bmatrix} 1 & 0 & x_q \\ 0 & 1 & y_q \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_{\text{scale}} = \begin{bmatrix} s_q & 0 & 0 \\ 0 & s_q & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ &\mathbf{T}_{\text{trans}}^{-1} = \begin{bmatrix} 1 & 0 & -x_q \\ 0 & 1 & -y_q \\ 0 & 0 & 1 \end{bmatrix}, \qquad (q = 1, 2, 3). \end{aligned} \tag{4}$$

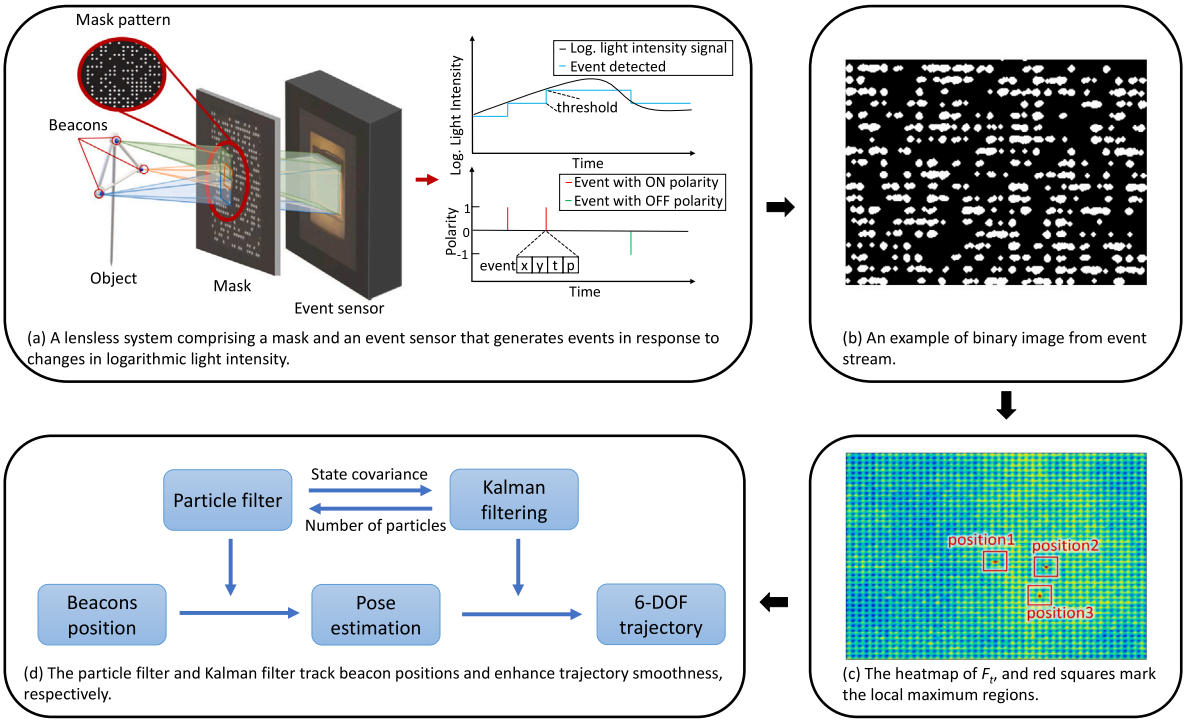


Fig. 2. The flowchart of 6-DOF object tracking method.

Here, $(x_q, y_q) \in \{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}$ in Eq. (3), $(i, j) \in [x_q : x_q + m - 1, y_q : y_q + n - 1]$, and $m \times n$ represents the spatial resolution of the sensor, with its dimensions aligned with I_s. The $\mathbf{T}_{\text{trans}}$ is the translation matrix that shifts the coordinate system origin to the target point (x_q, y_q), enabling the scaling to be performed relative to this specific location. The $\mathbf{T}_{\text{scale}}$ denotes the scaling matrix, which enlarges or reduces the region around (x_q, y_q) by the factor s_q while preserving its relative position within the global coordinate system. Finally, the $\mathbf{T}_{\text{trans}}^{-1}$ translates the coordinate system back to its original position, ensuring that the scaled region is correctly aligned in the global image space.

The I_{w1}, I_{w2}, and I_{w3} are overlapped, and the result is compared with I_s. The non-similarity is quantified using the following loss function:

$$F_{loss} = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \left(I_s[i,j] - \left(I_{w1}[i,j] \vee I_{w2}[i,j] \vee I_{w3}[i,j] \right) \right)^2. \tag{5}$$

The parameters x_q, y_q, and s_q are optimized using the gradient descent method:

$$\begin{aligned} (x_q, y_q, s_q)_{t+1} &= (x_q, y_q, s_q)_t \\ &\quad - \alpha \cdot (\nabla_{x_q} F_{\text{loss}}, \nabla_{y_q} F_{\text{loss}}, \nabla_{s_q} F_{\text{loss}}), \\ \text{where} \quad \nabla_{x_q} F_{\text{loss}} &= \frac{\partial F_{\text{loss}}}{\partial x_q}, \\ \nabla_{y_q} F_{\text{loss}} &= \frac{\partial F_{\text{loss}}}{\partial y_q}, \\ \nabla_{s_q} F_{\text{loss}} &= \frac{\partial F_{\text{loss}}}{\partial s_q}, \\ & (q = 1, 2, 3). \end{aligned} \tag{6}$$

Here, $(x_q, y_q, s_q)_t$ represents the parameter value after t iterations, α is the learning rate, and ∇L_t represents the gradient of the loss function with respect to the parameters. In each iteration step, the parameters are updated in order to minimize the loss function, thus to obtain the precise projection positions of the three beacons in the mask pattern, with the default number of iterations set to 10.

3.3. Pose estimation and motion filtering

To minimize noise interference and enhance the smoothness of the tracking trajectory, our method introduces a dual-filter approach in pose estimation, as shown in Fig. 2(d). The approach builds upon established hybrid filtering strategies in tracking systems [30], where the complementary strengths of different filters are exploited. Specifically, we employ PF to continuously monitor the projection positions of three beacons. Particle filtering improves beacon position accuracy through multiple sampling and weighted averaging, thereby mitigating the impact of marker extraction errors on the final pose estimation, similar to approaches used in [31,32]. Based on these positions, the Perspective-Three-Point (P3P) algorithm [33] estimates the object pose, which is then refined using UKF. The UKF effectively smooths the pose trajectory by compensating for motion dynamics and measurement uncertainties, as demonstrated in [34,35]. The parameters of PF and UKF are co-optimized to ensure consistent and coherent state estimation in our dual-filter framework.

The state vector of PF is augmented by concatenating the beacon state vectors as follows:

$$\mathbf{x} = \left[\mathbf{x}_{T1}^\top \ \mathbf{x}_{T2}^\top \ \mathbf{x}_{T3}^\top \right]^\top, \tag{7}$$

where the qth registration vector $\mathbf{x}_{Tq}$ represents the projection position (x_q, y_q) and velocity V_q of the qth beacon, as follows:

$$\mathbf{x}_{Tq} = \left[x_q \ y_q \ V_q \right]^\top, \quad (q = 1, 2, 3). \tag{8}$$

A set of N particles is initialized, each assigned equal weights, with the initial number of particles set to 200. Subsequently, a prediction is made using the state transition function f_p,

$$\hat{\mathbf{p}}_i = f_p(\mathbf{p}_i) + \mathbf{v}, \tag{9}$$

where $\mathbf{v} \sim \mathcal{N}(0, \Sigma_{\mathbf{v}})$ represents the noise added to the state transition, $\mathbf{p}_i$ is the state vector of the ith particle at the previous time step, and $\hat{\mathbf{p}}_i$ is the current state vector. Here, we assume that the position and velocity of the beacons keeps constant within a very short time period. Thus, the state transition function $f_p(*)$ is set to the identity.

The measurement vector obtained from the sensor is as follows:

$$\mathbf{z} = \left[\mathbf{z}_{T1}^{\top}\ \mathbf{z}_{T2}^{\top}\ \mathbf{z}_{T3}^{\top}\right]^{\top}, \quad \text{where} \quad \mathbf{z}_{Tq} = \left[x_q\ y_q\right]^{\top}, \quad (q = 1, 2, 3). \tag{10}$$

The likelihood of each particle is computed based on $\mathbf{z}$. The particles' weights are updated and particles are resampled to prioritize high-weight ones. The estimated state vector $\hat{\mathbf{x}}$ and its covariance matrix σ are calculated as follows:

$$\hat{\mathbf{x}} = \sum_{i=1}^{N} w_i \hat{\mathbf{p}}_i, \qquad \sigma = \sum_{i=1}^{N} w_i (\hat{\mathbf{p}}_i - \hat{\mathbf{x}})(\hat{\mathbf{p}}_i - \hat{\mathbf{x}})^{\top}, \tag{11}$$

where w_i is the weight of the ith particle.

Based on the projection positions $\hat{\mathbf{x}}$ and beacons' positional relationships in 3D space, the P3P algorithm [33] can be used to determine the object's pose as follows:

$$\mathbf{X} = \mathrm{P3P}(\hat{\mathbf{x}}, \mathbf{K}, \mathbf{d}), \tag{12}$$

where $\mathbf{K}$ is the internal matrix of system, $\mathbf{d}$ is the positional relationship among the three beacons. The output $\mathbf{X} = (x,\ y,\ z,\ \phi,\ \theta,\ \psi)^{\top}$, where $(x,\ y,\ z)^{\top}$ represents the position of the object, while $(\phi,\ \theta,\ \psi)^{\top}$ indicates the roll, pitch, and yaw angles of the object. It is noted that the beacons are subject to a constraint where they must be oriented towards the sensor. This constraint disambiguates at potential solutions provided by the P3P algorithm, ensuring a more accurate estimation of the object's pose.

The UKF is used to refine the object pose, and its state vector is defined as $\mathbf{X}$. The sigma point selection formula is as follows:

$$\chi = \{\mathbf{X},\ \mathbf{X} \pm \sqrt{(n+1)\mathbf{P}} \cdot \alpha\}, \tag{13}$$

where $n = 6$ is the dimension of the state vector, $\mathbf{P}$ is the state covariance matrix, and α represents a scaling factor that is associated with σ, as

$$\alpha = 1 - a_1 \cdot e^{-b_1 \cdot \mathbf{tr}(\sigma)}, \tag{14}$$

where $a_1,\ b_1 \in \mathbb{R}^{+}$, they determine the benchmark value for α and the accompanying σ controls the speed of increase. The posterior state estimation $\hat{\mathbf{X}}$ and posterior state covariance matrix $\hat{\mathbf{P}}$ can be obtained as follows:

$$\hat{\mathbf{X}} = \sum_{i=0}^{2n} w_i^m \chi_i + \mathbf{L}(\mathbf{Z} - \mathbf{H} \sum_{i=0}^{2n} w_i^m \chi_i), \qquad \hat{\mathbf{P}} = (\mathbf{I} - \mathbf{L}\mathbf{H}) \sum_{i=0}^{2n} w_i^c (\chi_i - \mathbf{X}^{-})(\chi_i - \mathbf{X}^{-})^{\top} + \mathbf{R}, \tag{15}$$

where $\mathbf{Z}$ is the observation vector from sensor, and $\mathbf{H}$ is the observation model, which can map $\mathbf{X}$ to $\hat{\mathbf{x}}$, thus representing the inverse operation of the P3P algorithm, i.e. $\mathbf{H} = \mathrm{P3P}^{-1}$. w^m is the mean weight, w^c is the covariance weight, $\mathbf{R} \sim \mathcal{N}(0, \Sigma_{\mathbf{R}})$ is the observation noise used to describe the uncertainty in the sensor measurement process, and $\mathbf{L}$ is the Kalman gain, used to avoid the accumulation of errors. $\hat{\mathbf{P}}$ is used to adjust the number of particles N for the next time of PF:

$$\beta = 1 - a_2 \cdot e^{-b_2 \cdot \mathbf{tr}(\hat{\mathbf{P}})}, \qquad N_{new} = N_{old} \cdot \beta, \tag{16}$$

where $a_2,\ b_2 \in \mathbb{R}^{+}$, analogous to the parameters in Eq. (14).

3.4. Calibration

This system adheres to the principles of pinhole camera models. In order to accurately track beacons, it is need to determine the intrinsic parameter $\mathbf{K}$ of the system, as

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{17}$$

where (c_x, c_y) represents the coordinates of the principal point, which is the projection of the sensor's center onto the mask pattern along the optical axis. Additionally, f_x and f_y represent the focal lengths in the x and y directions in the image coordinate system. These values are defined as the product of the distance between the mask plate and the sensor, the pixel size of the sensor, and a scaling factor. Although the imaging characteristics of lensless systems differ from those of traditional lens-based systems, the calibration method remains universally applicable because the proposed lensless system also adheres to the pinhole imaging model. Zhang's method [36] is employed to calibrate $\mathbf{K}$. Localization of three rigidly constrained beacons is required for this calibration. In principle, only two sets of data from distinct locations should suffice, but multiple rounds of data acquisition are often conducted to enhance precision. According to Liu et al. [37], the spherical motion model provides a more advanced option for improving calibration accuracy, which may be considered in future work.

4. Experimental analysis

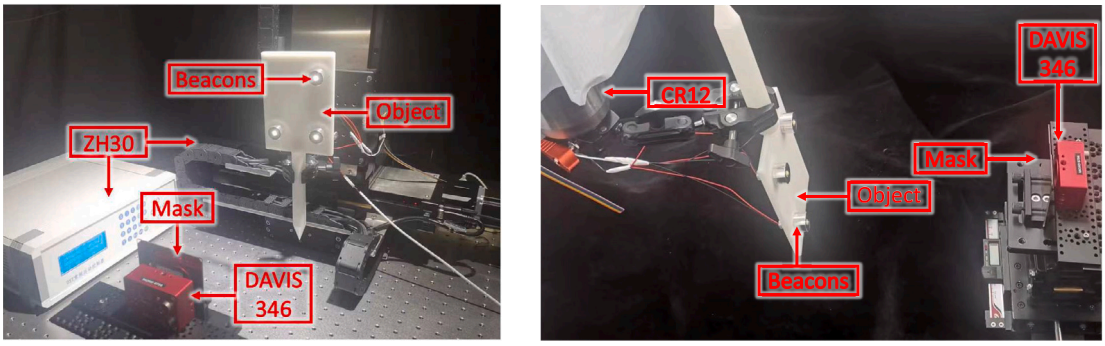
4.1. Experiment setup

The experiment setup is depicted in Fig. 3. Three beacons with rigid constraints are affixed to the object, which is mounted on an electric 3D platform (ZH30, KM) and a robotic arm (CR12, ROKAE) to assess our system's accuracy in the position and rotation estimations. Since the event sensor only responds to changes in light intensity, we make the beacons flash to generate events. The mask board (30.6 mm × 30.6 mm) is composed of photolithographic glass and printed with 128 × 128 DBT dot matrix pattern, as Section 3.1.1, with apertures of size 0.12 mm and spaced at the interval of 0.24 mm. The event sensor (DAVIS346, iniVation) has the spatial resolution of 346 × 260 pixels and it is operated using its default settings. All data are collected using a Windows PC equipped with a 2.9 GHz Intel Core i7-10700 processor.

4.2. Position and orientation RMSE evaluation

The RMSE (root mean square error) evaluation allow us to statistically determine the accuracy and consistency of the method. We track the position and rotation of the object, with the true trajectory values obtained separately from the electric 3D platform and the robotic arm. The object's trajectory is depicted in Fig. 4. Our method has an RMSE of 1.35 cm in positional motion and 3.08 degrees in rotational motion. Considering the sensitivity of event sensors to changes in light intensity, tracking objects during lighting variations presents a challenge. We demonstrate the performance by inducing lighting changes through toggling the lights in the laboratory. Remarkably, there is no significant decline in tracking accuracy when the laboratory lights are turned off and on, as illustrated by the gray shading in Fig. 4.

Additionally, since the algorithm utilizes beacons for object tracking, we conducted a comparative analysis with the methods of Ebmer et al. [7], which also employs beacons based on event sensors. Both methods are evaluated in the same environment, resulting in position and rotation errors of 1.68 cm and 2.94 degrees, respectively. In Ebmer's method, 8 beacons are used to form a marker, and rotational data is obtained from 4 markers, resulting in a significantly higher number of beacons compared to our system, which contributes to their lower rotational error. Nevertheless, our algorithm demonstrated comparable rotational accuracy, showing no notable disadvantage. Despite utilizing more beacons, the Ebmer's method exhibited a position error 19.6% higher than ours. This highlights the effectiveness of our lens-free system, particularly during translational motion, where the projection pattern undergoes higher frequency changes, thereby improving positional accuracy.



(a) (b)

Fig. 3. Experiment setup: (a) Position estimation of the object with the ground truth value from the electric 3D platform. (b) Rotation estimation of the object with the ground truth value from the robotic arm.

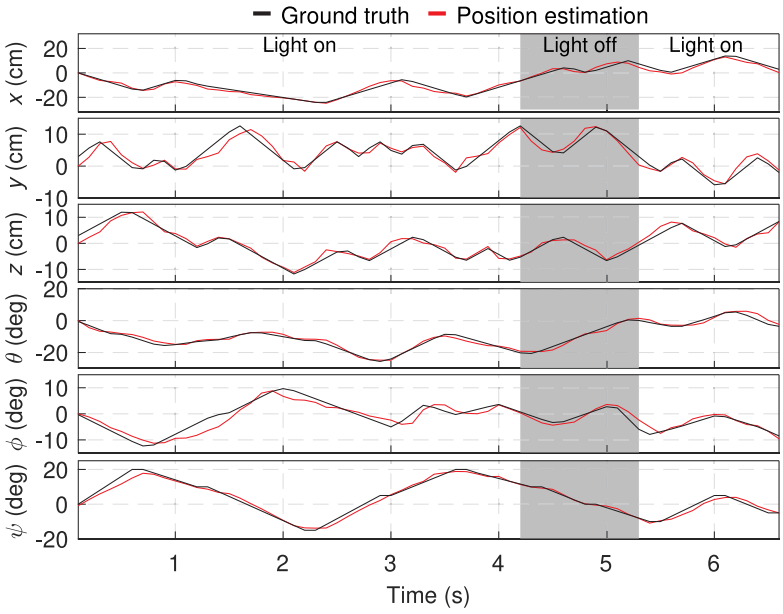


Fig. 4. 6-DOF motion trajectory of the object.

4.3. Multi object tracking

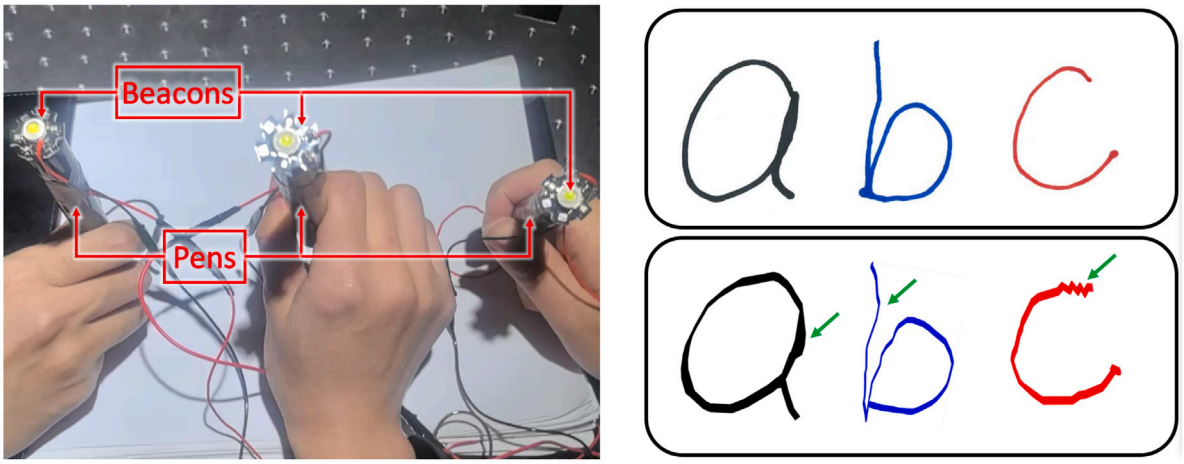
To evaluate the robustness of the proposed method in multi-object tracking scenarios, we conduct an experiment with three beacons attached to separate pens, which are simultaneously controlled to perform writing motions, as shown in Fig. 5(a). The trajectories of the pens' movements and the corresponding tracking results are comparatively illustrated in Fig. 5(b). Experimental results demonstrate that the method achieves robust long-term tracking performance in multi-object scenarios, even under challenging conditions involving complex motion trajectories.

As shown in Fig. 5(b), the tracked trajectory aligns with the hand-written trajectory in overall shape, although local discrepancies are observed. These discrepancies arise from two factors: **1. Geometric offset.** The beacon is mounted at the rear end of the pen, whereas handwriting motion is generated at the pen tip. When the pen is tilted, a natural geometric deviation occurs between the motion paths of the pen tip and the rear-mounted beacon, resulting in asynchronous movement. **2. Dynamic shaking.** At the end of writing the character 'a', the pen-lifting motion induces inertia, causing a slight rebound in the trajectory of the pen's rear beacon. At the beginning of characters 'b' and 'c', an increase in friction at the pen tip leads to unstable hand force, introducing brief shaking in the beacon trajectory (as indicated by the green arrow in Fig. 5(b)). Despite these localized differences, the overall tracking performance remains unaffected. All pen trajectories maintain correct target identity without switching, and the topological structure of the characters is accurately preserved.

4.4. Outdoor drone tracking

To validate the pose estimation performance of the proposed system in complex scenarios, we conduct outdoor experiments under uncontrollable lighting and environmental conditions. During the experiments, the beacons are mounted on a drone (FanciSwarm), while the lensless system is vertically fixed on a ground-based tripod, as shown in Fig. 6(a). The drone is programmed to perform a series of dynamic maneuvers, including rapid ascents, descents, and lateral movements, to rigorously test the system's capabilities.

The experimental results demonstrate that the system is capable of stably capturing and accurately reconstructing the drone's motion trajectory over a 30-second flight duration, as shown in Fig. 6(b). The reconstruction accuracy is evaluated by calculating the RMSE between the reconstructed trajectory and the ground truth. The resulting value of 1.41 cm demonstrates the high accuracy achieved by the method. To further illustrate the system's high-precision attitude estimation performance, we focus on a representative 10-second segment of the flight,



(a) (b)

Fig. 5. (a) Three beacons attached to pens with different colors, used for simultaneous writing on paper. (b) Top: Handwriting produced by pens. Bottom: Tracking results of beacons using the proposed method.

during which the drone exhibits complex maneuvers. The corresponding attitude estimation results (roll, pitch, and yaw) are depicted in Fig. 6(c). A quantitative evaluation of orientation accuracy is conducted using RMSE, resulting in 1.89 degrees for roll, 1.52 degrees for pitch, and 1.02 degrees for yaw. These findings fully underscore the system's superior performance in dynamic object tracking tasks, particularly in challenging outdoor environments.

4.5. Real-time performance

The real-time performance of tracking algorithms is crucial, as it represents the algorithm's ability to quickly process input data and update tracking results. The proposed algorithm accumulates the event stream to generate binary images and performs tracking based on these images. The duration of event accumulation is defined as the AP (Accumulation Period), which determines the update frequency of the algorithm, expressed as $1/AP$. The average time to generate and process a binary image is defined as the PP (Processing Period). The AP/PP ratio serves as a metric, where $AP/PP > 1$ indicates that the algorithm can process tracking tasks in real time.

Background tasks and the operating system are disabled to prevent unintentional interruptions or delays during the tracking process, resulting in a PP of 1.71 ms. However, since the processing time of individual images may exceed this average, we set the AP to 2 ms and the tracking frequency to 500 Hz to ensure the system's stability. By maintaining $AP/PP > 1$, the algorithm can meet real-time requirements in most cases while ensuring stable operation at 500 Hz.

4.6. Ablation experiment

To evaluate the trade-off between accuracy and real-time performance introduced by dual-filtering, we conduct ablation experiments based on the setup in Fig. 4, as summarized in Table 1. Removing the dual-filtering module increases the localization error by 61.3% but improves the processing period by only 0.58 ms. It is noteworthy that the filtering process accounts for only 33.9% of the total system time, as the majority of the computational load lies in the localization stage based on beacon projections. This confirms the effectiveness of sacrificing a small amount of computational efficiency to achieve higher accuracy.

In the scalability evaluation with respect to the number of beacons, when the number of beacons increased from 3 to 5, the system, benefiting from enhanced observation constraints, automatically reduced the number of particles due to the dual-filtering design. As a result, the processing period stabilizes at 1.94 ms without a significant increase, while the pose error is further reduced by 6.5%. These results demonstrate that the proposed dual-filtering strategy maintains a favorable balance between accuracy and efficiency, even as the system scales.

4.7. Limitations and performance evaluation

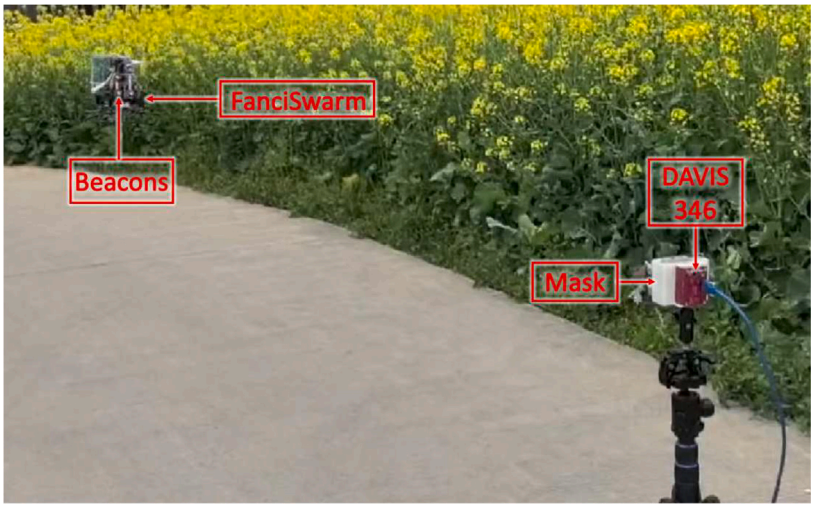
This section presents a detailed analysis of the proposed method, examining its limitations and performance. Through experiments, we evaluate the impact of object depth, beacons angle, and object speed on tracking accuracy.

4.7.1. Object depth

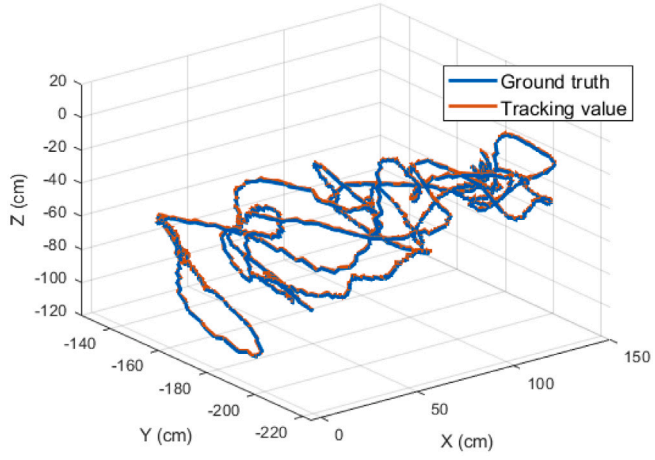
In the experiment, we assess the influence of varying depths on accuracy by positioning the object plane at different distances from the sensor plane. The object is fixed on the electric 3D platform and moves 10 cm along the X-axis. The results indicate that the proposed method achieves the highest accuracy when the object depth is about 45 cm, as shown in Fig. 7(a). As the depth increased, the beacons signal strength detected by the event sensor diminishes, and background noise rose, leading to a decrease in tracking accuracy. While enhancing the brightness of beacons or increasing the intensity of spatial filtering could mitigate this issue, we maintain experimental consistency by not implementing such adjustments. Conversely, when the object depth is insufficient, fewer faculae are detected, reducing the effectiveness of overdetermination and consequently lowering accuracy.

4.7.2. Beacons angle

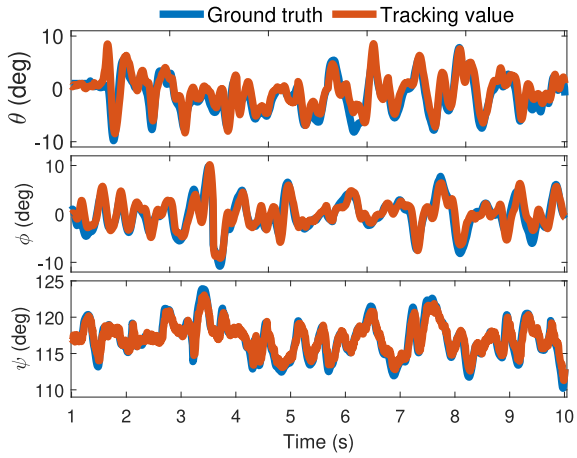
In practical applications, variations in the angle between sensors and beacons can result in different projection effects, thereby affecting tracking accuracy. By fixing the beacons and sensor at different relative angles and moving the object 10 cm along the Y-axis, we evaluate the effect of the beacons' incidence angle on tracking accuracy. The results indicate that when the incidence angle is within $\pm 20°$, it leads to minimal image distortion and higher tracking accuracy, as shown in Fig. 7(b). Beyond $\pm 20°$, increased image distortion leads to a decline in tracking performance, potentially requiring more complex mask pattern and algorithms, which in turn increases hardware and computational costs. However, since most tracked objects do not experience extreme angle shifts relative to the sensor, this range is common in applications.



(a)

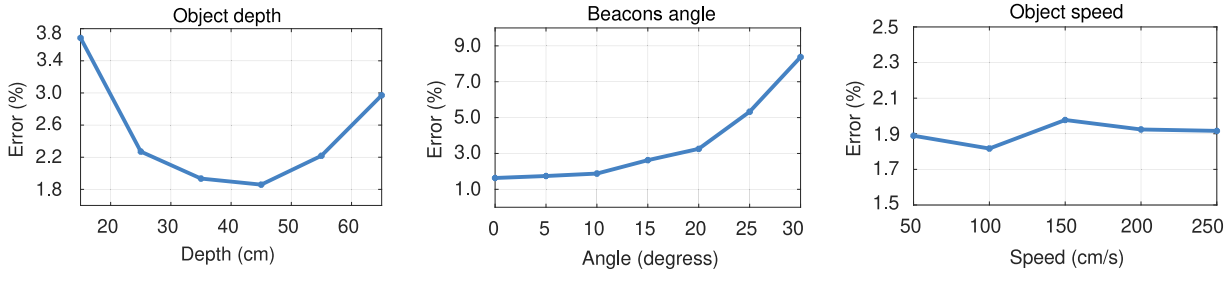


(b)



(c)

Fig. 6. (a) Schematic diagram of the outdoor drone tracking experimental setup using lensless imaging. (b) Reconstructed motion trajectory of the drone. (c) Attitude estimation results.



(a) (b) (c)

Fig. 7. Relationship between object depth, beacon angle, object velocity, and tracking accuracy.

Table 1
Impact of dual filtering and beacon quantity on localization error and processing period.

Parameter	Original configuration	Without filtering	Increased beacons
Error (cm)	1.68	2.71	1.57
Processing period (ms)	1.71	1.13	1.94

4.7.3. Object speed

To assess the proposed method's performance in tracking high-speed objects, the object is moved along the X-axis at a constant speed. We define the object speed range as 50 to 250 cm/s, which corresponds to fast motion for event sensors, where objects traversing the field of view in 0.2 to 0.9 s are typically considered fast. The results indicate that object speed has minimal impact on accuracy, as shown in Fig. 7(c). At lower speeds, the beacon's flashing allows the event sensor to generate enough events for tracking. Even at higher speeds, the tracking frequency of 500 Hz is sufficient to maintain high-precision tracking, demonstrating the method's broad applicability.

5. Conclusion

In this paper, an event-based 6-DOF object tracking method is proposed. The use of event sensors improves the sampling rate, enabling the method to achieve 500 Hz tracking frequency to enhance the stability of capturing fast-moving objects. The application of masks compensates for the low spatial resolution of event sensors, allowing the method to achieve higher accuracy. Experimental results show that ours achieved a position RMSE of only 1.35 cm and a rotation RMSE of only 3.08 degrees, significantly surpassing the accuracy of other event-based trackers. Additionally, the method requires low computational resources, making it suitable for real-time applications, and robust to ambient light changes. The current system is primarily designed for moderate-scale tracking scenarios. However, large-scale applications present potential challenges, including the limited field of view of event sensors and constraints in beacon configuration. Moreover, when beacons projections overlap heavily, the robustness of the deconvolution algorithm may be challenged, resulting in reduced localization accuracy. Future work will focus on extending the system's applicability by incorporating sensor arrays or optimizing beacon deployment strategies. We also plan to develop event trackers utilizing multi-layer masks to further enhance the accuracy of rotation estimation. In addition, we envision generalizing the framework to more complex scenarios, such as tracking non-rigid objects or operation in non-laboratory environments, to broaden its practical applicability.

CRediT authorship contribution statement

Yue You: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Mingzhu Zhu:** Writing – review & editing, Methodology, Formal analysis. **Hong Zhang:** Conceptualization. **Bingwei He:** Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

[1] C. Taylor, M. Evans, E. Crellin, M. Parsons, D. Cosker, Ego-interaction: Visual hand-object pose correction for VR experiences, in: Proc. ACM SIGGRAPH Conf. Motion, Interact. Games, CMIG, 2021, pp. 1–8.
[2] W.A. El Ahmar, D. Kolhatkar, F.E. Nowruzi, H. AlGhamdi, J. Hou, R. Laganiere, Multiple object detection and tracking in the thermal spectrum, in: Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit., CVPR, 2022, pp. 277–285.
[3] X. Wang, F. Li, Q. Du, Y. Zhang, T. Wang, G. Fu, C. Lu, Micro-amplitude vibration measurement using vision-based magnification and tracking, Meas. 208 (2023) 112464.
[4] L. Chen, L. Ma, W. Zhan, Y. Zhang, L. Sun, Research on the method of anti-occlusion of surgical instrument tracking based on multi-camera module information fusion, Meas. 239 (2025) 115480.
[5] P. Lichtensteiner, C. Posch, T. Delbruck, A 128x128 120db 15μs latency asynchronous temporal contrast vision sensor, IEEE J. Solid-State Circuits (2) (2008) 566–576.
[6] Z. Li, N.A. Piga, F. Di Pietro, M. Iacono, A. Glover, L. Natale, C. Bartolozzi, Hybrid object tracking with events and frames, in: Proc. 2023 IEEE/RSJ Int. Conf. Intell. Robots Syst., IROS, IEEE, 2023, pp. 9057–9064.
[7] G. Ebmer, A. Loch, M.N. Vu, R. Mecca, G. Haessig, C. Hartl-Nesic, M. Vincze, A. Kugi, Real-time 6-dof pose estimation by an event-based camera using active LED markers, in: Proc. IEEE/CVF Winter Conf. Appl. Comput. Vision, 2024, pp. 8137–8146.
[8] W. Jin, M. Zhu, J. Liu, B. He, J. Yu, Shadow-based lightsource localization with direct camera-lightsource geometry, IEEE Trans. Instrum. Meas. (2023).
[9] Z. Wang, Y. Ng, J. Henderson, R. Mahony, Smart visual beacons with asynchronous optical communications using event cameras, in: 2022 IEEE/RSJ Int. Conf. Intell. Robots Syst., IROS, IEEE, 2022, pp. 3793–3799.
[10] Z. Xunt, J. Huang, Z. Li, Z. Ying, Y. Wang, C. Xu, F. Gao, Y. Cao, Crepes: Cooperative relative pose estimation system, in: 2023 IEEE/RSJ Int. Conf. Intell. Robots Syst., IROS, IEEE, 2023, pp. 5274–5281.
[11] M. Zhu, B. He, J. Yu, F. Yuan, J. Liu, Hydramarker: efficient, flexible, and multifold marker field generation, IEEE Trans. Pattern Anal. Mach. Intell. 45 (5) (2022) 5849–5861.
[12] J. Carpenter, P. Clifford, P. Fearnhead, Improved particle filter for nonlinear problems, IEE Proc.-Radar Sonar Navig. 146 (1) (1999) 2–7.
[13] E.A. Wan, R. Van Der Merwe, The unscented Kalman filter for nonlinear estimation, in: Proc. IEEE 2000 Adaptive Syst. Signal Process., Commun., Control Symp., SPCC, IEEE, 2000, pp. 153–158.
[14] M. Aladem, S.A. Rawashdeh, A multi-cluster tracking algorithm with an event camera, in: Proc. 2019 IEEE Nat. Aerosp. Electronics Conf., NAECON, IEEE, 2019, pp. 391–397.
[15] A. Dietsche, G. Cioffi, J. Hidalgo-Carrió, D. Scaramuzza, Powerline tracking with event cameras, in: Proc. 2021 IEEE/RSJ Int. Conf. Intell. Robots Syst., IROS, IEEE, 2021, pp. 6990–6997.
[16] Z. Liu, B. Guan, Y. Shang, Q. Yu, L. Kneip, Line-based 6-DoF object pose estimation and tracking with an event camera, IEEE Trans. Image Process. (2024).
[17] Z. Liu, B. Guan, Y. Shang, S. Liang, Z. Yu, Q. Yu, Optical flow-guided 6dof object pose tracking with an event camera, in: Proc. 32nd ACM Int. Conf. Multimedia, 2024, pp. 6501–6509.
[18] Z. Liu, B. Guan, Y. Shang, Y. Bian, P. Sun, Q. Yu, Stereo event-based, 6-DOF pose tracking for uncooperative spacecraft, IEEE Trans. Geosci. Remote Sens. (2025).
[19] A. Mitrokhin, C. Fermüller, C. Parameshwara, Y. Aloimonos, Event-based moving object detection and tracking, in: IEEE/RSJ Int. Conf. Intell. Robots Syst., IROS, IEEE, 2018, pp. 1–9.
[20] S. Li, Z. Zhou, Z. Xue, Y. Li, S. Du, Y. Gao, 3D feature tracking via event camera, in: Proc. IEEE/CVF Conf. Computer Vision Pattern Recognit., 2024, pp. 18974–18983.
[21] X. Cai, Z. You, H. Zhang, J. Gu, W. Liu, T. Xue, Phocolens: Photorealistic and consistent reconstruction in lensless imaging, Adv. Neural Inf. Process. Syst. 37 (2024) 12219–12242.
[22] Y. Zhou, G. Yan, W. Guo, Y. Yang, X. Guan, Artifacts-free lensless on-chip tomography empowered by three-dimensional deconvolution, Appl. Phys. Lett. 125 (6) (2024).
[23] N.G. De Bruijn, A combinatorial problem, Proc. Sect. Sci. K. Ned. Akad. Wet. Amst. 49 (7) (1946) 758–764.
[24] D. Schüsselbauer, A. Schmid, R. Wimmer, Dothraki: Tracking tangibles atop tabletops through de-bruijn tori, in: Proc. Int. Conf. Tangible, Embedded, Embodied Interact., TEEI, 2021, pp. 1–10.
[25] Y. Feng, H. Lv, H. Liu, Y. Zhang, Y. Xiao, C. Han, Event density based denoising method for dynamic vision sensor, Appl. Sci. 10 (6) (2020) 2024.
[26] T. Huang, B. Zheng, J. Zhang, C. Yi, Y. Jiang, Q. Shui, H. Jian, Mango surface defect detection based on HALCON, in: Adv. Inf. Technol. Electron. Automat. Control Conf., vol. 5, IEEE, 2021, pp. 2627–2631.
[27] Z. Gong, L. Su, H. Liu, Y. Gu, M. Yue, Transfer learning-based image processing of recognizing office objects for indoor mobile robots, in: Int. Conf. Adv. Robot. Mechatronics, IEEE, 2023, pp. 31–36.

[28] D. Kundur, D. Hatzinakos, Blind image deconvolution, IEEE Signal Process. Mag. 13 (3) (2002) 43–64.
[29] A.K. Eldaly, Y. Altmann, A. Perperidis, S. McLaughlin, Deconvolution of irregularly subsampled images, in: 2018 IEEE Statistical Signal Process. Workshop, IEEE, 2018, pp. 303–307.
[30] G. Cao, D. Li, N. Zhou, D. Li, et al., Pedestrian and vehicle trajectory prediction method based on YOLO and hybrid filtering., J. Eng. Sci. Technol. Rev. 18 (3) (2025).
[31] X. Wu, J. Yang, Track-before-detect algorithm-based optimized interactive multiple model particle filtering for tracking maneuvering weak target, in: Int. Conf. Image Process. Artif. Intell., vol. 13213, SPIE, 2024, pp. 606–613.
[32] N. Zhou, L. Lau, R. Bai, T. Moore, A genetic optimization resampling based particle filtering algorithm for indoor target tracking, Remote. Sens. 13 (1) (2021) 132.
[33] M. Persson, K. Nordberg, Lambda twist: An accurate fast robust perspective three point (p3p) solver, in: Proc. Eur. Conf. Comput. Vision, ECCV, 2018, pp. 318–332.
[34] Y. Zhang, M. Li, Y. Zhang, Z. Hu, Q. Sun, B. Lu, An enhanced adaptive unscented Kalman filter for vehicle state estimation, IEEE Trans. Instrum. Meas. 71 (2022) 1–12.
[35] T.H. Park, S. D'Amico, Adaptive neural-network-based unscented kalman filter for robust pose tracking of noncooperative spacecraft, J. Guid. Control Dyn. 46 (9) (2023) 1671–1688.
[36] Z. Zhang, A flexible new technique for camera calibration, IEEE Trans. Pattern Anal. Mach. Intell. 22 (11) (2000) 1330–1334.
[37] Z. Liu, S. Liang, B. Guan, D. Tan, Y. Shang, Q. Yu, Collimator-assisted high-precision calibration method for event cameras, Optim. Lett. 50 (13) (2025) 4254–4257.